

Epidemiology and pathophysiology of alcohol and breast cancer: Update 2012.

AIMS: To update epidemiological data on alcohol and breast cancer, with special emphasis on light alcohol consumption, and to review mechanisms of alcohol mediated mammary carcinogenesis. **METHODS:** For epidemiological data, in November 2011 we performed a literature search in various bibliographic databases, and we conducted a meta-analysis of data on light alcohol drinking. Relevant mechanistic studies were also reviewed to November 2011. **RESULTS:** A significant increase of the order of 4% in the risk of breast cancer is already present at intakes of up to one alcoholic drink/day. Heavy alcohol consumption, defined as three or more drinks/day, is associated with an increased risk by 40-50%. This translates into up to 5% of breast cancers attributable to alcohol in northern Europe and North America for a total of approximately 50,000 alcohol-attributable cases of breast cancer worldwide. Up to 1-2% of breast cancers in Europe and North America are attributable to light drinking alone, given its larger prevalence in most female populations when compared with heavy drinking. Alcohol increases estrogen levels, and estrogens may exert its carcinogenic effect on breast tissue either via the ER or directly. Other mechanisms may include acetaldehyde, oxidative stress, epigenetic changes due to a disturbed methyl transfer and decreased retinoic acid concentrations associated with an altered cell cycle. **CONCLUSIONS:** Women should not exceed one drink/day, and women at elevated risk for breast cancer should avoid alcohol or consume alcohol occasionally only.